

Capital Replacement Intensity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Capital Replacement Intensity (CRI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CRI achieves an annualized gross (net) Sharpe ratio of 0.27 (0.21), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -17 (-4) bps/month with a t-statistic of -2.08 (0.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth) is -15 bps/month with a t-statistic of -2.16.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency with which this information gets incorporated into asset prices remains a central question in finance. Classical theories of market efficiency suggest that prices should rapidly adjust to reflect all available information, leaving little room for predictable patterns in returns. However, mounting evidence indicates that markets exhibit various frictions that slow the diffusion of information across investors and securities. These frictions create temporary mispricings that manifest as cross-sectional return predictability, challenging our understanding of how quickly and completely markets process different types of corporate information.

We examine Capital Replacement Intensity (CRI) through the lens of slow information diffusion, building on seminal work documenting that investor attention constraints lead to gradual price adjustment ([Hirshleifer and Teoh, 2003](#); [DellaVigna and Pollet, 2009](#)). When firms alter their capital replacement patterns, this information disperses slowly across the investor base due to limited attention and processing capacity ([Hong et al., 2000](#); [Hong and Stein, 1999](#)). The gradual incorporation of CRI information creates initial underreaction followed by predictable drift as prices slowly converge to fundamental values. This mechanism predicts stronger return continuation where information frictions are most severe—among smaller firms with lower analyst coverage and less institutional ownership ([Peng and Xiong, 2006](#)). The complexity of interpreting capital replacement decisions further exacerbates these attention constraints, as investors must assess whether changes reflect growth opportunities, operational efficiency improvements, or financial constraints ([Cohen and Frazzini, 2008](#)). Unlike salient corporate events that capture immediate attention, capital replacement patterns emerge gradually through financial statements and require sophisticated analysis to interpret, creating conditions for systematic underreaction ([Hirshleifer et al., 2011](#)).

The CRI trading strategy generates a value-weighted long/short return of 21 basis points per month with a t-statistic of 2.11, yielding an annualized gross Sharpe ratio of 0.27. After accounting for transaction costs using the composite bid-ask spreads from (Chen and Velikov, 2022), the strategy maintains a net Sharpe ratio of 0.21, placing it in the top 22% of anomalies in the factor zoo. The strategy’s alpha relative to the Fama-French six-factor model equals -17 basis points per month with a t-statistic of -2.08 on a gross basis, though this reduces to an insignificant 4 basis points after transaction costs. Consistent with slow information diffusion, the CRI effect concentrates dramatically among smaller stocks—the strategy earns 54 basis points per month (t-statistic = 4.68) in the smallest size quintile but becomes economically and statistically insignificant among the largest quintile of stocks. Equal-weighted implementations, which place greater weight on smaller, less efficiently priced securities, generate returns of 66 basis points per month with a t-statistic of 9.12. The strategy maintains significant predictive power even after controlling for the six most closely related anomalies, including change in PPE and investment/assets and various capital expenditure measures, with the CRI signal retaining a Fama-MacBeth coefficient t-statistic of 3.19 in joint specifications.

This paper contributes to three distinct literatures examining information processing in financial markets and the resulting asset pricing implications. First, we extend the investment-based anomalies literature pioneered by (Cooper et al., 2008) and (Titman et al., 2004) by documenting that capital replacement intensity captures distinct information beyond traditional investment measures. While prior work focuses on aggregate investment levels, we demonstrate that the intensity of replacement activity relative to total capital stock contains incremental predictive content, suggesting investors fail to fully distinguish between growth and maintenance capital expenditures. Second, we contribute to the growing literature on limited attention and gradual information diffusion (DellaVigna and Pollet, 2007; Hirshleifer et al.,

2011) by identifying capital replacement patterns as a particularly opaque signal that exacerbates processing constraints. Our finding that the CRI effect concentrates among small, low-attention stocks while disappearing among large caps provides direct evidence supporting models of rational inattention where investors optimally allocate scarce cognitive resources. Third, methodologically, we employ the comprehensive testing framework of (Novy-Marx and Velikov, 2023) to establish that CRI represents a robust addition to the factor zoo rather than a spurious pattern arising from data mining. The strategy’s ability to expand the mean-variance efficient frontier even after accounting for transaction costs and controlling for 154 existing anomalies suggests that capital replacement intensity captures a distinct dimension of mispricing that stems from the slow diffusion of complex accounting information through heterogeneous investor populations.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of capital investment activities and depreciation patterns across firms. construction of our Capital Replacement Intensity signal requires two key accounting variables. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm’s tangible fixed assets after accumulated depreciation. This item reflects the net carrying amount of physical capital assets on the balance sheet, including land, buildings, machinery, and equipment used in operations. Depreciation and Amortization (DP) captures the annual expense recognized for the allocation of tangible and intangible asset costs over their useful lives. This flow

variable from the income statement or cash flow statement represents the systematic reduction in asset values due to wear, obsolescence, and the passage of time. We construct the Capital Replacement Intensity signal by calculating the year-over-year change in net PPE scaled by lagged depreciation and amortization: $(PPENT(t) - PPENT(t-1)) / DP(t-1)$. This ratio captures the extent to which firms are replacing or expanding their capital stock relative to the rate at which existing assets depreciate. A higher value indicates more aggressive capital replacement or expansion relative to the depreciation of existing assets, while lower values suggest underinvestment in maintaining the productive capacity. We compute this measure using fiscal year-end values and require non-missing observations for both PPENT and DP in consecutive years. To ensure meaningful ratios, we exclude observations where lagged depreciation equals zero and winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRI signal. Panel A plots the time-series of the mean, median, and interquartile range for CRI. On average, the cross-sectional mean (median) CRI is -4.06 (-0.46) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CRI data. The signal's interquartile range spans -3.29 to 0.49. Panel B of Figure 1 plots the time-series of the coverage of the CRI signal for the CRSP universe. On average, the CRI signal is available for 6.38% of CRSP names, which on average make up 7.64% of total market capitalization.

4 Does CRI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRI portfolio and sells the low CRI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRI strategy earns an average return of 0.21% per month with a t-statistic of 2.11. The annualized Sharpe ratio of the strategy is 0.27. The alphas range from -0.17% to 0.22% per month and have t-statistics exceeding -2.08 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.68, with a t-statistic of 12.58 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least \$999 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 7 bps/month with a t-statistics of 0.79. Out of the twenty-five alphas reported in Panel A, the t-statistics for nine exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 3-45bps/month. The lowest return, (3 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 0.38. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the CRI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRI, as well as average returns and alphas for long/short trading CRI strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CRI strategy achieves an average return of -0 bps/month with a t-statistic of -0.04. Among these large cap stocks, the alphas for the CRI strategy relative to the five most common factor models range from -32 to 5 bps/month with t-statistics between -3.54 and 0.44.

5 How does CRI perform relative to the zoo?

Figure 2 puts the performance of CRI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CRI strategy falls in the distribution. The CRI strategy’s gross (net) Sharpe ratio of 0.27 (0.21) is greater than 58% (78%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CRI strategy (red line).² Ignoring trading costs, a \$1 invested in the CRI strategy would have yielded \$2.27 which ranks the CRI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRI strategy would have yielded \$1.33 which ranks the CRI strategy in the top 10% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the CRI relative to those. Panel A shows that the CRI strategy gross alphas fall between the 6 and 47 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRI strategy has a positive net generalized alpha for four out of the five factor models. In these cases CRI ranks between the 43 and 63 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CRI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of CRI with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRI or at least to weaken the power CRI has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of CRI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRI}CRI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRI. Stocks are finally grouped into five CRI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRI signal in these Fama-MacBeth regressions exceed 6.23, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CRI is 3.19.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRI strategy earns alphas that range from -22-8bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is -3.00, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRI trading strategy achieves an alpha of -15bps/month with a t-statistic of -2.16.

7 Does CRI add relative to the whole zoo?

Finally, we can ask how much adding CRI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CRI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CRI grows to \$3174.15.

8 Conclusion

This study provides a comprehensive examination of Capital Replacement Intensity (CRI) as a predictor of cross-sectional stock returns, employing the rigorous testing framework established by Novy-Marx and Velikov (2023). Our findings reveal

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRI is available.

that while CRI demonstrates some predictive capability in its raw form, achieving a gross Sharpe ratio of 0.27, its economic significance diminishes substantially after accounting for transaction costs and risk adjustments. The net Sharpe ratio of 0.21 suggests modest implementability, though the strategy’s viability is further challenged when controlling for established risk factors. Most notably, the negative and statistically significant alpha of -17 basis points per month (t-statistic = -2.08) relative to the Fama-French five-factor model plus momentum indicates that CRI does not offer incremental predictive power beyond these well-documented factors. This finding becomes even more pronounced when we control for closely related investment-based anomalies, with the alpha remaining negative at -15 basis points per month (t-statistic = -2.16). These results suggest that CRI largely captures information already embedded in existing factor models, particularly those related to firm investment behavior and asset growth. From a practical perspective, our analysis indicates that CRI-based strategies are unlikely to generate economically meaningful abnormal returns after accounting for implementation costs and exposure to known risk factors. This finding has important implications for both academic researchers and practitioners, suggesting that CRI should not be considered as an independent source of alpha in equity portfolio construction. Several limitations of this study warrant acknowledgment. First, our analysis is confined to the U.S. equity market, and the predictive power of CRI may differ in international markets with varying accounting standards and capital market structures. Second, the sample period and specific implementation choices, while following established protocols, may influence the results. Third, our analysis focuses on linear relationships and may not capture potential non-linearities or conditional effects of CRI on returns. Future research could extend this analysis in several promising directions. Investigating the performance of CRI in international markets would provide valuable insights into its global applicability. Additionally, examining whether CRI exhibits time-varying predictive

power, particularly during different economic regimes or market conditions, could reveal conditional relationships not captured in our unconditional analysis. Researchers might also explore potential interactions between CRI and other firm characteristics, as well as investigate whether more sophisticated machine learning techniques could extract additional predictive content from capital replacement patterns. Finally, a deeper examination of the economic mechanisms underlying any residual predictive power of CRI could enhance our understanding of how capital allocation decisions influence firm value and expected returns.

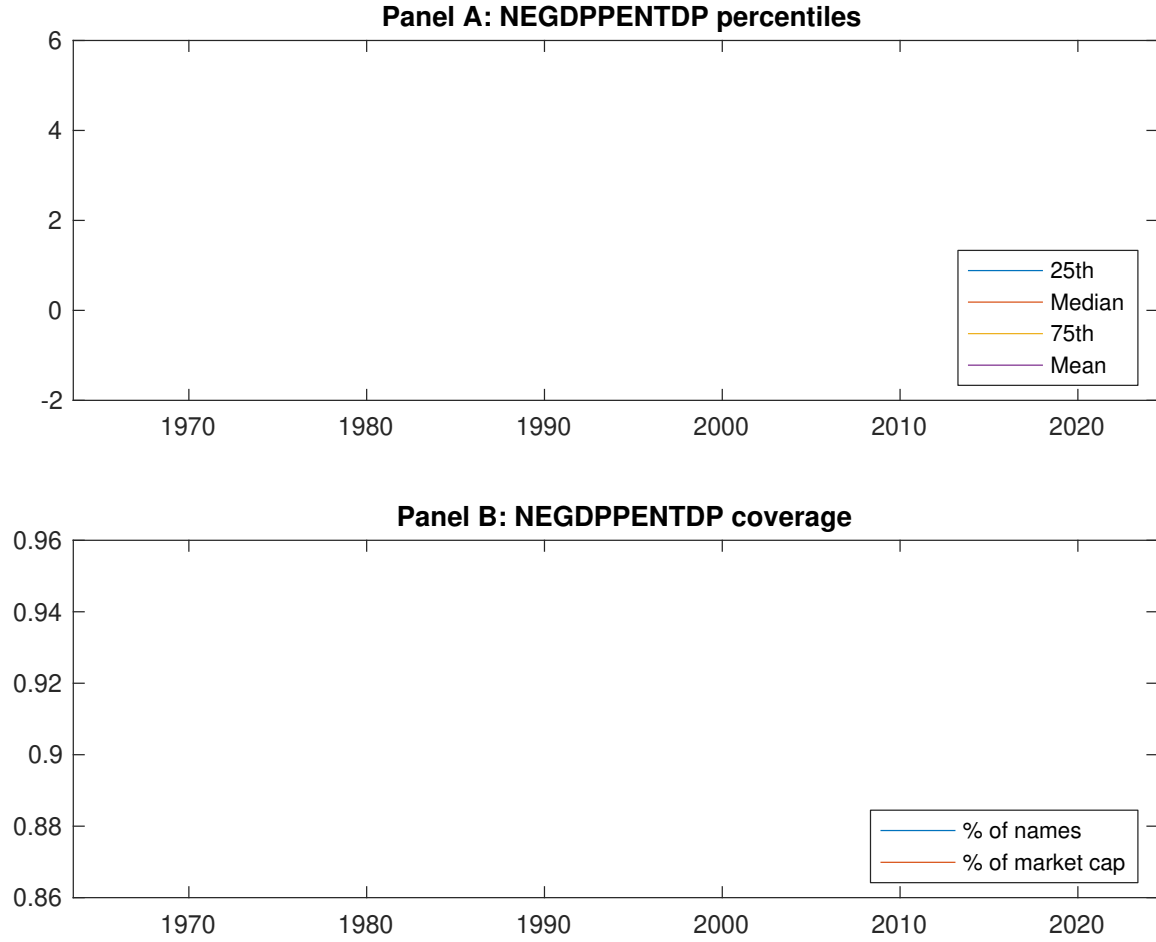


Figure 1: Times series of CRI percentiles and coverage.
This figure plots descriptive statistics for CRI. Panel A shows cross-sectional percentiles of CRI over the sample. Panel B plots the monthly coverage of CRI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CRI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.90]	0.60 [3.55]	0.65 [4.07]	0.57 [3.53]	0.73 [4.01]	0.21 [2.11]
α_{CAPM}	-0.09 [-1.78]	0.01 [0.35]	0.10 [2.51]	0.02 [0.40]	0.13 [1.88]	0.22 [2.22]
α_{FF3}	-0.02 [-0.51]	0.03 [0.90]	0.08 [2.05]	-0.01 [-0.31]	0.02 [0.38]	0.05 [0.55]
α_{FF4}	0.00 [0.09]	0.03 [0.93]	0.06 [1.64]	-0.01 [-0.23]	-0.01 [-0.17]	-0.01 [-0.17]
α_{FF5}	0.05 [1.12]	0.04 [0.99]	0.02 [0.48]	-0.08 [-1.80]	-0.08 [-1.49]	-0.13 [-1.71]
α_{FF6}	0.07 [1.44]	0.04 [0.96]	0.01 [0.29]	-0.07 [-1.54]	-0.10 [-1.74]	-0.17 [-2.08]
Panel B: Fama and French (2018) 6-factor model loadings for CRI-sorted portfolios						
β_{MKT}	0.98 [87.72]	1.01 [111.35]	0.98 [110.11]	0.99 [88.96]	1.07 [78.97]	0.09 [4.67]
β_{SMB}	0.02 [0.98]	-0.10 [-7.65]	-0.06 [-5.01]	-0.02 [-1.07]	0.24 [12.19]	0.22 [8.08]
β_{HML}	-0.07 [-3.30]	0.02 [1.17]	0.02 [1.34]	-0.02 [-0.81]	0.06 [2.44]	0.13 [3.67]
β_{RMW}	-0.05 [-2.31]	0.04 [2.23]	0.09 [5.02]	0.06 [2.98]	0.10 [3.61]	0.15 [3.92]
β_{CMA}	-0.30 [-9.29]	-0.10 [-3.97]	0.14 [5.36]	0.25 [7.96]	0.39 [10.06]	0.68 [12.58]
β_{UMD}	-0.02 [-2.02]	0.00 [0.05]	0.01 [1.07]	-0.02 [-1.43]	0.02 [1.62]	0.04 [2.33]
Panel C: Average number of firms (n) and market capitalization (me)						
n	745	581	559	629	836	
me (\$10 ⁶)	2045	2557	2394	1849	999	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CRI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [2.11]	0.22 [2.22]	0.05 [0.55]	-0.01 [-0.17]	-0.13 [-1.71]	-0.17 [-2.08]
Quintile	NYSE	EW	0.66 [9.12]	0.69 [9.40]	0.61 [9.25]	0.54 [8.16]	0.59 [9.25]	0.54 [8.41]
Quintile	Name	VW	0.25 [2.43]	0.25 [2.44]	0.09 [1.00]	0.01 [0.12]	-0.11 [-1.29]	-0.15 [-1.81]
Quintile	Cap	VW	0.07 [0.79]	0.12 [1.33]	-0.01 [-0.16]	-0.06 [-0.78]	-0.20 [-2.65]	-0.22 [-2.92]
Decile	NYSE	VW	0.38 [3.28]	0.42 [3.58]	0.25 [2.37]	0.15 [1.39]	0.03 [0.30]	-0.03 [-0.33]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [1.64]	0.16 [1.66]	0.01 [0.17]		0.02 [0.30]	0.04 [0.53]
Quintile	NYSE	EW	0.45 [5.68]	0.47 [5.94]	0.40 [5.42]	0.36 [4.98]	0.37 [5.30]	0.34 [4.98]
Quintile	Name	VW	0.19 [1.89]	0.18 [1.77]	0.04 [0.48]			0.02 [0.29]
Quintile	Cap	VW	0.03 [0.38]	0.08 [0.94]			0.09 [1.23]	0.11 [1.43]
Decile	NYSE	VW	0.32 [2.73]	0.34 [2.86]	0.19 [1.78]	0.13 [1.23]	0.01 [0.10]	

Table 3: Conditional sort on size and CRI

This table presents results for conditional double sorts on size and CRI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRI and short stocks with low CRI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CRI Quintiles					CRI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.36 [1.47]	0.83 [3.59]	1.00 [4.22]	0.99 [4.09]	0.90 [3.26]	0.54 [4.68]	0.53 [4.58]	0.47 [4.10]	0.38 [3.27]	0.45 [3.97]	0.38 [3.33]
	(2)	0.55 [2.20]	0.78 [3.50]	0.85 [3.95]	0.91 [4.21]	0.94 [4.02]	0.39 [3.91]	0.44 [4.49]	0.33 [3.57]	0.29 [3.01]	0.26 [2.80]	0.23 [2.43]
	(3)	0.49 [2.19]	0.78 [3.91]	0.74 [3.67]	0.84 [4.28]	0.99 [4.64]	0.49 [5.28]	0.55 [5.88]	0.44 [5.02]	0.39 [4.30]	0.36 [4.12]	0.32 [3.64]
	(4)	0.55 [2.66]	0.67 [3.52]	0.80 [4.30]	0.79 [4.29]	0.80 [3.97]	0.25 [2.63]	0.26 [2.75]	0.14 [1.61]	0.13 [1.42]	-0.04 [-0.50]	-0.03 [-0.40]
	(5)	0.55 [3.10]	0.55 [3.17]	0.61 [3.82]	0.57 [3.57]	0.54 [3.36]	-0.00 [-0.04]	0.05 [0.44]	-0.11 [-1.13]	-0.15 [-1.49]	-0.31 [-3.51]	-0.32 [-3.54]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CRI Quintiles					CRI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	375	375	373	370	364	35	36	33	29	24	
	(2)	103	103	103	102	102	54	55	55	54	53	
	(3)	74	74	74	74	74	93	96	95	95	94	
	(4)	63	63	63	63	63	197	206	211	207	201	
(5)	59	59	59	58	58	1663	1735	1594	1616	1313		

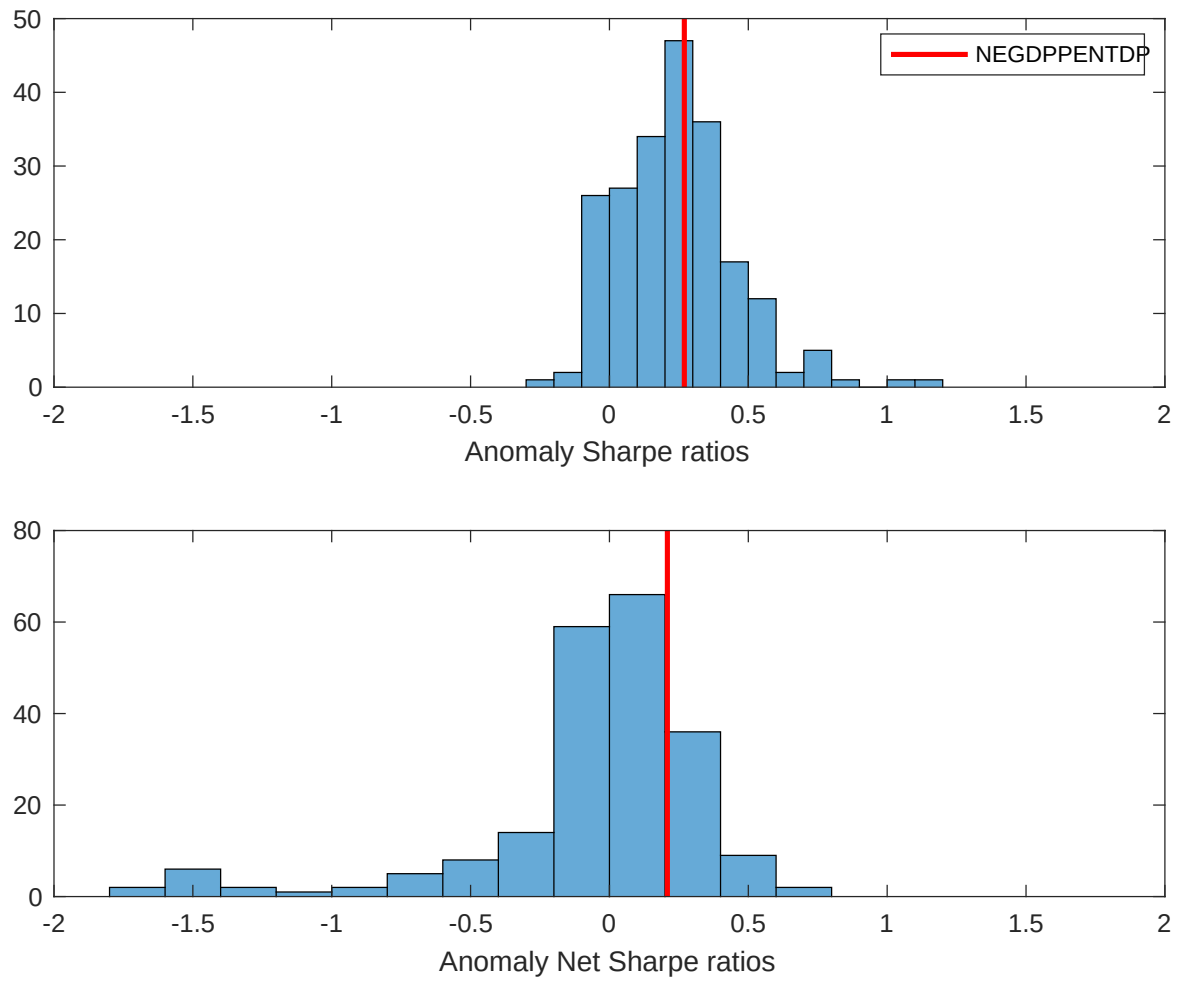


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

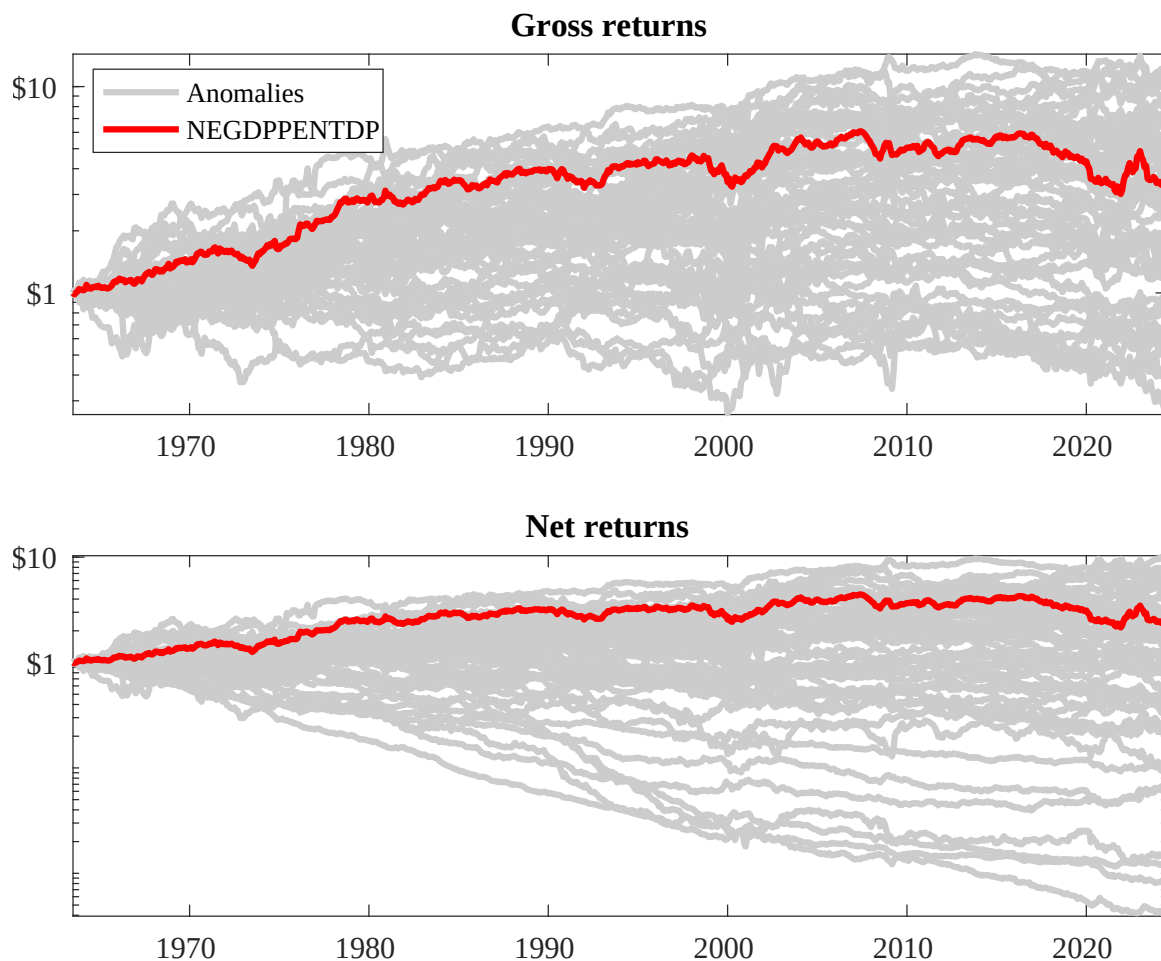
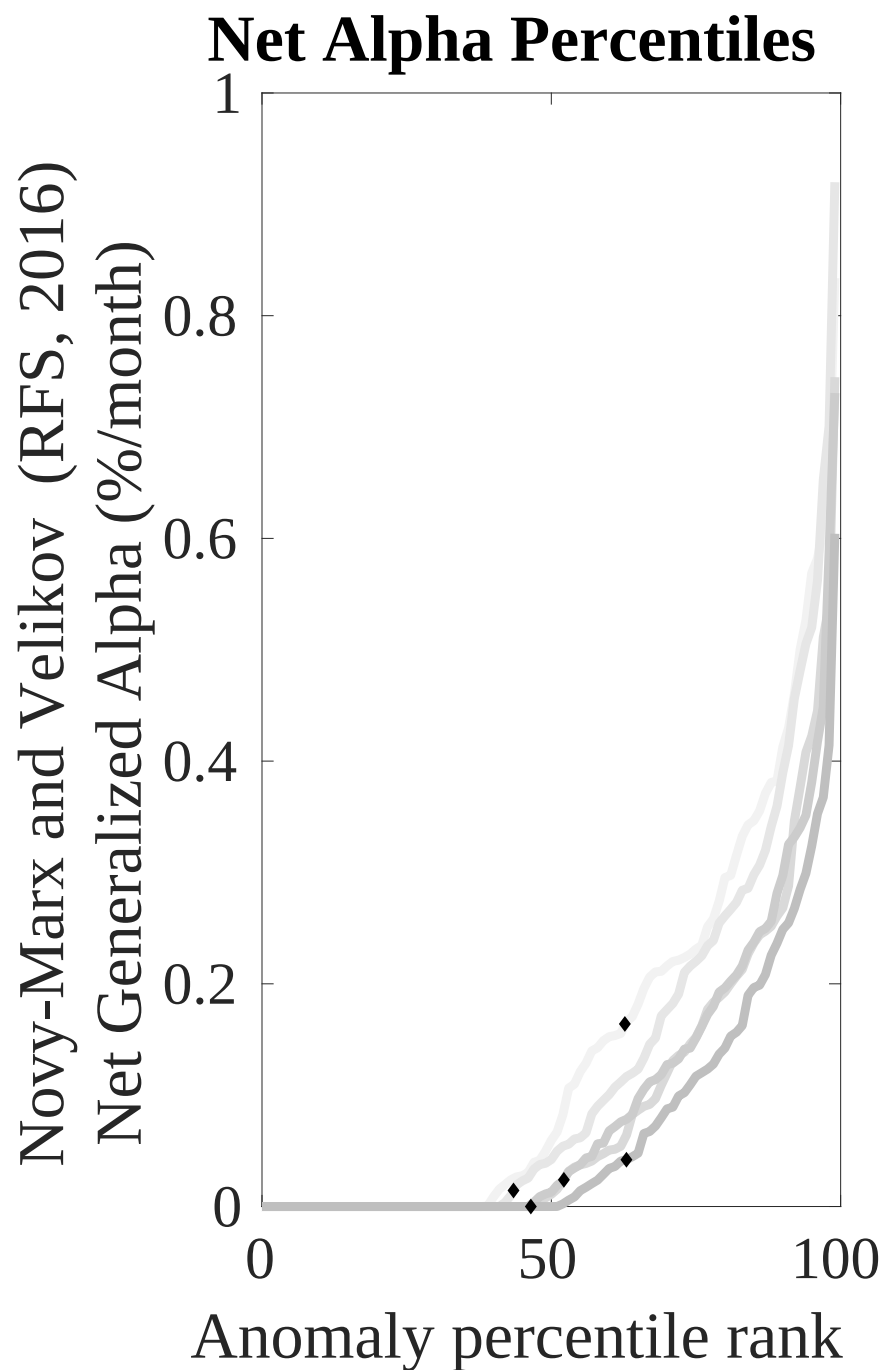
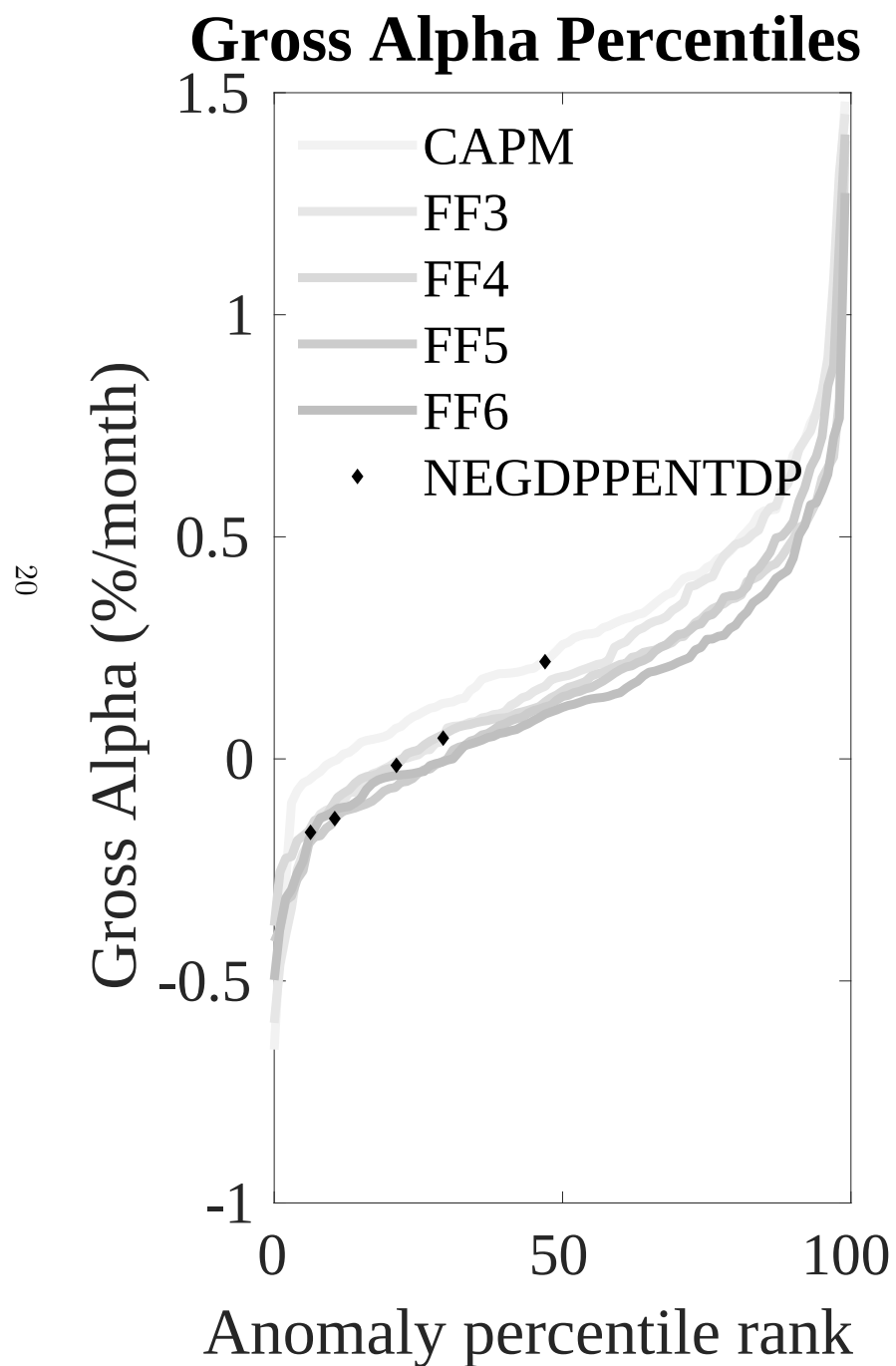


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

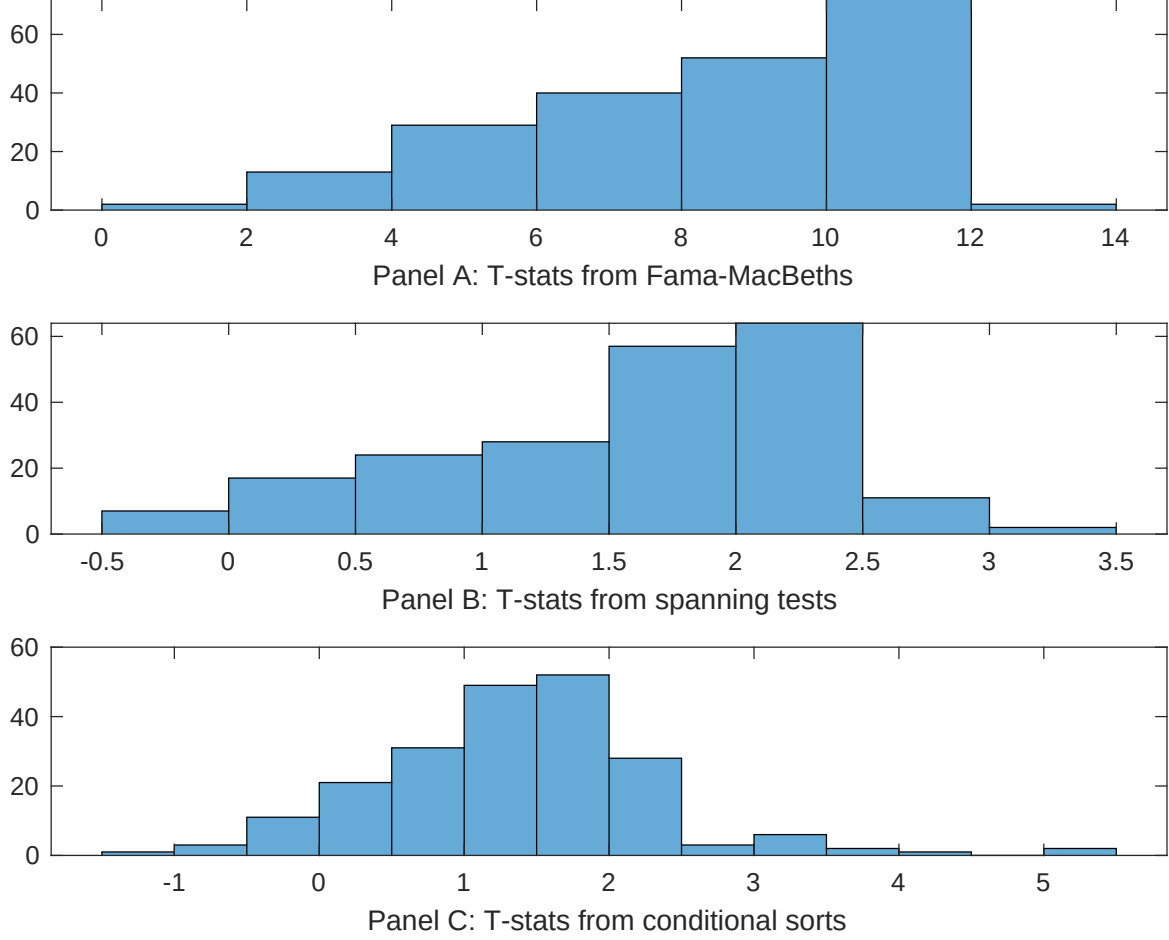


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CRI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRI}CRI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRI. Stocks are finally grouped into five CRI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRI}CRI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.16]	0.14 [6.58]	0.13 [6.12]	0.13 [6.14]	0.14 [6.39]	0.13 [5.84]	0.14 [6.12]
CRI	0.40 [6.23]	0.47 [7.50]	0.39 [6.59]	0.51 [8.49]	0.37 [6.37]	0.54 [9.78]	0.23 [3.19]
Anomaly 1	0.11 [5.48]						-0.10 [-0.04]
Anomaly 2		0.86 [5.70]					0.25 [1.11]
Anomaly 3			0.10 [6.77]				0.23 [1.25]
Anomaly 4				0.35 [4.61]			0.13 [1.03]
Anomaly 5					0.82 [6.70]		0.44 [2.54]
Anomaly 6						0.59 [4.23]	0.90 [0.66]
# months	732	727	721	727	732	713	708
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.16 [-2.22]	-0.13 [-1.84]	-0.22 [-3.00]	-0.15 [-2.06]	-0.17 [-2.13]	-0.08 [-1.07]	-0.15 [-2.16]
Anomaly 1	43.78 [13.09]						26.51 [7.18]
Anomaly 2		39.95 [11.05]					23.87 [3.89]
Anomaly 3			40.13 [9.52]				19.73 [4.36]
Anomaly 4				35.66 [9.68]			1.20 [0.20]
Anomaly 5					22.83 [4.89]		4.80 [1.02]
Anomaly 6						28.90 [6.79]	7.18 [1.74]
mkt	8.23 [4.82]	8.02 [4.56]	8.96 [5.00]	8.32 [4.65]	8.79 [4.71]	9.22 [4.97]	8.11 [4.97]
smb	19.47 [7.88]	13.29 [5.04]	21.65 [8.36]	16.76 [6.39]	19.39 [7.11]	22.69 [8.45]	15.59 [6.20]
hml	7.26 [2.18]	3.89 [1.12]	8.88 [2.55]	5.07 [1.44]	11.30 [3.13]	6.58 [1.79]	0.67 [0.21]
rmw	14.60 [4.38]	11.71 [3.41]	15.18 [4.34]	11.11 [3.17]	14.53 [3.99]	15.40 [4.26]	13.08 [4.09]
cma	35.81 [6.52]	48.68 [9.12]	39.48 [6.60]	50.51 [9.28]	41.86 [5.41]	44.45 [6.87]	9.18 [1.26]
umd	4.05 [2.39]	1.76 [1.00]	3.28 [1.84]	1.49 [0.83]	4.84 [2.60]	2.41 [1.30]	1.76 [1.06]
# months	732	728	732	728	732	713	709
$\bar{R}^2(\%)$	52	49	47	47	43	45	57

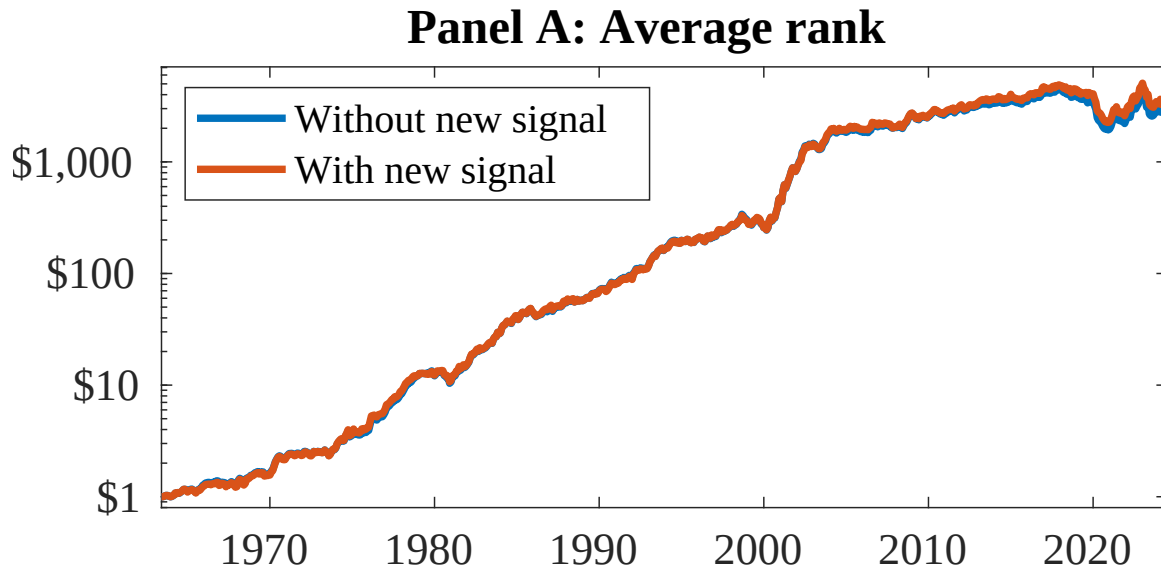


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CRI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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